

closure_evidence_20260923

BOB-234 — closure evidence (2026-09-23)

Field	Value
Revision	1
Last modified	2026-09-23T00:00:00Z
Item	BOB-234 (Bug)
Evidence class	runtime (live merge service :7187 + download-proxy :7186, token-armed)

No token value appears anywhere in this file or in the changed sources (§11.4.10).

Root cause

tests/integration/test_merge_api.py used a bare requests.Session. The live merge service runs with BOBA_API_TOKEN armed (BOB-197), and download-proxy/src/api/routes.py::require_api_token answers 401 unless the request carries Authorization: Bearer <token> or X-Boba-Token: <token>.

Second defect found while fixing (same root cause, previously masked because the test never got past the 401): the download test's cleanup POST :7186/api/v2/torrents/delete was also unauthenticated. The :7186 proxy (plugins/download_proxy.py) gates deletes with the same token, so the cleanup got 401. Its response was never checked, and the test torrent was left in qBittorrent. Observed live: after the first green

run, one itest-74e5b1dd torrent (tags Boba, 5o6a) remained. An unauthenticated delete returned 401. The same delete with the token returned 200 and the torrent was removed.

Fix (test-side only)

- NEW tests/integration/merge_api_auth.py:
 - resolve_api_token() reads the environment first, then the repo .env.
 - MergeTokenSession sends X-Boba-Token only to the merge-service origin (scheme+host+port match).
 - assert_token_usable() probes POST /api/v1/magnet. It FAILS loudly (never skips) when the service is armed and the token is missing, or when the token is rejected.
 - qbit_delete_confirmed() sends the token to the :7186 delete, asserts HTTP 200 and polls until the hash is gone.
- tests/integration/test_merge_api.py: new api_token fixture; session is now a MergeTokenSession; the download cleanup uses qbit_delete_confirmed.
- NEW tests/unit/test_bob234_merge_api_sends_token.py: 14 tests against a real local http.server (401 without the header, 200 with it).

RED -> GREEN

- RED (unit): collection failed with ModuleNotFoundError: tests.integration.merge_api_auth. The cleanup tests then failed (2 failed / 12 passed) before qbit_delete_confirmed existed.
- Pre-fix live behaviour: 10 x HTTP 401, recorded in docs/Issues.md BOB-234.
- Paired mutation (session reverted to a bare requests.Session, then restored with cmp-verified byte identity): 8 failed (hooks x5, magnet x3).
- Wrong-token negative control (BOBA_API_TOKEN=<wrong> in env): the fixture fails with “rejected the BOBA_API_TOKEN taken from environment (POST /api/v1/magnet -> HTTP 401)”. It does not skip.
- GREEN: tests/unit/test_bob234_merge_api_sends_token.py + tests/integration/test_merge_api.py gave 34 passed in 73.86s.

Debris (CM-NO-TEST-TAG-DEBRIS / §11.4.14)

```
PRE torrents 0 tags ['1080p', 'Boba', 'Бoбa'] boba-hyphen-tags []
hooks 0 itest_scripts 0
POST torrents 0 tags ['1080p', 'Boba', 'Бoбa'] boba-hyphen-tags []
hooks 0 itest_scripts 0
CM-NO-TEST-TAG-DEBRIS: PASS — no 'boba-' test debris among 3 live
tag(s)
```

The torrent left by the first (pre-cleanup-fix) run was removed manually with an authenticated delete. The final run left none.

Honest gaps

- The production tags Boba/Бoбa were already in qBittorrent's tag list before the final run. They are production tags, not boba- debris. The test does not delete tags (deleting shared production tags could affect real torrents).
- Other integration files that call token-gated routes were not audited (out of scope). They may have the same 401 latent defect.

Addendum — diff-scoped cleanup (CM-NO-UNSCOPED-LIVE-DESTRUCTION)

The first commit attempt was refused by invariant CM-NO-UNSCOPED-LIVE-DESTRUCTION because `qbit_delete_confirmed` deleted by hash with no baseline. Reworked to the gate's pattern: `qbit_snapshot_hashes` is taken BEFORE the add; the delete runs only when the hash is in after - before; an unreadable baseline (None) disarms the delete. Evidence (this session):

- `pytest tests/unit/test_bob234_merge_api_sends_token.py -> 17 passed.`
- Mutation (drop the `before is None` guard and the `h not in added` guard): 2 failed
(..._disarmed_when_baseline_unreadable, ..._never_deletes_a_preexisting_torrent)
15 passed; restored -> 17 passed.
- `bash scripts/pre_build/check_cm_no_unscoped_live_destruction.sh -> PASS, 10 files, all scoped.`
- Live `pytest tests/integration/test_merge_api.py -> 20 passed;`
`check_cm_no_test_tag_debris.sh -> PASS.`